

Multi-Echelon Supply Chain Network Design via Mixed-Integer Linear Programming (MILP)

Mathematical Derivation and Optimization Formulation

1. Problem Definition

A global supply chain consists of factories, warehouses and customer zones. The objective is to decide facility-opening choices and shipment quantities while minimizing total operating expenditure.

2. Sets and Indices

Symbol	Meaning
$i \in F$	Factory index
$j \in W$	Warehouse index
$k \in C$	Customer-zone index

3. Parameters

Parameter	Meaning
S_i	Capacity of factory i
W_j	Capacity of warehouse j
D_k	Demand of customer zone k
c_{ij}	Unit cost from factory i to warehouse j
c_{jk}	Unit cost from warehouse j to customer k
f_i	Fixed factory-opening cost
g_j	Fixed warehouse-opening cost
α_k	Minimum service level for customer k
M	Large positive linking constant

4. Decision Variables

- x_{ij} = quantity shipped from factory i to warehouse j .
- y_{jk} = quantity shipped from warehouse j to customer zone k .
- s_k = quantity satisfying the required service condition.
- u_i = 1 if factory i is opened; 0 otherwise.
- v_j = 1 if warehouse j is opened; 0 otherwise.

5. Objective Function Derivation

Total cost = factory-to-warehouse transportation + warehouse-to-customer transportation + fixed factory costs + fixed warehouse costs.

Minimize $Z = \sum_i \sum_j c_{ij} x_{ij} + \sum_j \sum_k c_{jk} y_{jk} + \sum_i f_i u_i + \sum_j g_j v_j$

6. Factory Capacity Constraint

$\sum_j x_{ij} \leq S_i u_i \quad \forall i$

A closed factory has $u_i = 0$, so its shipment becomes zero. An opened factory can ship up to S_i units.

7. Warehouse Flow-Balance Constraint

$$\sum_i x_{ij} = \sum_k y_{kj} \quad \forall j$$

This ensures that warehouse outbound flow cannot exceed its inbound flow.

8. Warehouse Capacity Constraint

$$\sum_k y_{kj} \leq W_j v_j \quad \forall j$$

A closed warehouse cannot handle shipments, while an opened warehouse is limited by W_j .

9. Customer Demand Constraint

$$\sum_j y_{kj} = D_k \quad \forall k$$

Every customer zone receives its required demand.

10. Minimum Service-Level Constraint

$$\sum_j s_{kj} \geq \alpha_k D_k \quad \forall k$$

This guarantees that at least the required fraction of customer demand meets the specified service condition.

11. Linking Constraints

$$x_{ij} \leq M u_j \quad \forall i, j$$

$$y_{kj} \leq M v_j \quad \forall j, k$$

These constraints connect continuous shipment decisions to binary facility-opening decisions.

12. Variable Restrictions

$$x_{ij}, y_{kj}, s_{kj} \geq 0$$

$$u_j, v_j \in \{0,1\}$$

13. Matrix Form

Let $z = [x, y, s, u, v]$. The complete model can be written as:

Minimize $c^T z$
subject to $Az \leq b$
 $Ez = d$
 $z \geq 0$
 $u, v \in \{0,1\}$

Matrix A contains capacity, service-level and linking constraints. Matrix E contains flow-balance and demand equations.

14. Constraint Matrix Structure

Constraint group	Matrix form	Purpose
Factory capacity	$A_F z \leq b_F$	Limits factory flow
Warehouse capacity	$A_W z \leq b_W$	Limits warehouse capacity
Service level	$A_S z \geq b_S$	Guarantees minimum service
Flow balance	$E_B z = d_B$	Conserves material
Demand	$E_D z = d_D$	Satisfies customer demand
Facility linking	$A_L z \leq b_L$	Links flow and opening variables

15. Complete MILP Formulation

Minimize $Z = \sum_i \sum_j c_{ij} x_{ij} + \sum_k \sum_l c_{kl} y_{kl} + \sum_k f_k u_k + \sum_l g_l v_l$

Subject to:

$\sum_j x_{ij} \leq S_i u_i \quad \forall i$
 $\sum_i x_{ij} = \sum_l y_{lj} \quad \forall j$
 $\sum_l y_{lj} \leq W_j v_j \quad \forall j$
 $\sum_l y_{lj} = D_k \quad \forall k$
 $\sum_l s_{kl} \geq \alpha_k D_k \quad \forall k$
 $x_{ij} \leq M u_i \quad \forall i, j$
 $y_{jk} \leq M v_j \quad \forall j, k$
 $x_{ij}, y_{jk}, s_{kl} \geq 0$
 $u_i, v_j \in \{0,1\}$

16. Optimization Interpretation

The MILP solver selects the factories and warehouses to open and determines the shipment quantity on every transportation lane. The optimal solution minimizes total cost while satisfying capacity, flow balance, demand and service-level requirements.

17. Conclusion

The derived formulation converts the multi-echelon supply-chain network design problem into a formal Mixed-Integer Linear Program. It provides a systematic way to obtain a cost-efficient global logistics network under operational and customer-service constraints.